microCT imaging of threespine stickleback

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Abstract

Can we predict evolution? The three-spined stickleback (Gasterosteus aculeatus) is a well-recognized system for understanding adaptation to divergent habitats. Populations of benthic and limnetic stickleback differ in a number of phenotypic traits that are associated with shifts in dietary specialization. However, analyses of the structures required for feeding – especially the jaws and complex internal branchial anatomy – requires considerable time and expertise, with destructive sampling and fine dissection skills needed for quantitative analysis. The advent of µCT and 3D-scanning technology affords non-destructive sampling and an increase the resolution of data available for study, but at the substantial cost of increasing complexity and processing time for each specimen. To address these concerns, we developed a rapid and semi-automated segmentation and analysis pipeline based around the Biomedisa Image Segmentation platform for investigating three-dimensional morphological adaptation within the three-spined stickleback. The pipeline includes splitting 44 multi-specimen scans into regions of interest for each specimen, reconstruction of targeted anatomy, and morphometric analyses. We then applied this pipeline to a sampling effort encompassing hundreds of samples of divergent benthic and limnetic stickleback populations (N=216), showcasing the possibility of using high-throughput scanning data to provide tests of ecological and evolutionary hypotheses.

Introduction

- Embedded into Alaska Stickleback Restoration Project, Genomics axis where Katie Peichel, Ben Sulser and Sheila Christen are affiliated.
- Why are we studying what we are studying in these fish?

Micro-computed tomography

X-ray microtomography is an indispensable tool to gain non-destructive insights into the inner structure of highly diverse samples, namely for specimens studied in the biomedical sciences [1]. Microtomographic imaging is ideally suited to non-destructively assess the morphology of different fish species, large and small [2].

Materials & Methods

Sample procurement and preparation

Due to their inherent contrast difference to the surrounding tissue, the structures of interest we touch upon in this manuscript (teeth and bones, i.e. jaws and skull) are well visualized in unstained samples, hence no further preparation of the fish was necessary.

microtomographic imaging

In a small pilot study we determined the optimal scanning parameters to fit the constraints on total scanning time, resolution and sample handling. To optimize for these constraints, we scanned all the sticklebacks in batches of six fish in a custom-made 3D printed sample holder in a single scan. This holder was generated with OpenSCAD and is available online, either directly as STL file for printing or as (parameterized) OpenSCAD file for adaptation to other classes of samples. Both files are part of a library of 3D-printable sample holders for tomographic imaging [3].

Tomographic imaging was performed on a Bruker SkyScan 2214 at the Institute of Anatomy, University of Bern, Switzerland. In total we performed 44 scans, each of the scan usually containing 6 fish in the sample holder.

The relevant details of each scan are collated in a table in the Supplementary Materials; a short overview of the scanning parameters is given below. The X-ray source was set to a voltage of 60 kV and a current of around 110 µA for all but one scan where we used a source voltage of 49 kV and 159 µA due to operator error. For each sample, we recorded a set of 3601 projections of approximately 3000 x 2000 pixels at every 0.1° over a 360° sample rotation. Every single projection was exposed for about a second (depending on the sample). Due to the length of the fish, we had to acquire so-called stacked scans, on average we scanned 3 fields of view along the rotation axis of the sample holder. This resulted in scan times between 3 to 5 hours hours. The projection images were then subsequently reconstructed into a 3D stack of images with NRecon (Bruker microCT, Kontich Belgium, Version: 2.1.0.1 or 2.2.0.6). The isometric voxel sized in the resulting datasets vary from 15 to 19 µm.

Data analysis

Preparation and handling of tomographic datasets

After acquisition, a simple script was used to copy the relevant data to both archival storage and storage accessible by all co-authors at the same time.

Further processing of the tomographic dataset was performed with a set of Jupyter [4] notebooks [5]. The scripts are freely available online under the MIT License and may be freely used, modified, and redistributed for research, teaching, and other non-military purposes.

Preview notebook

The preview notebook is used for surfacing issues with the scanning. For this, we read all relevant scanning and reconstruction parameters from the log files of each scan. Afterwards, we efficiently loading the reconstruction PNG images from disk with the `dask_image.imread.imread` function [6]. Like so, we can map all the generated reconstructions to memory and quickly generate maximum intensity projections (MIP) of each scan (see Figure 1 for an example) for both quality control and further processing.

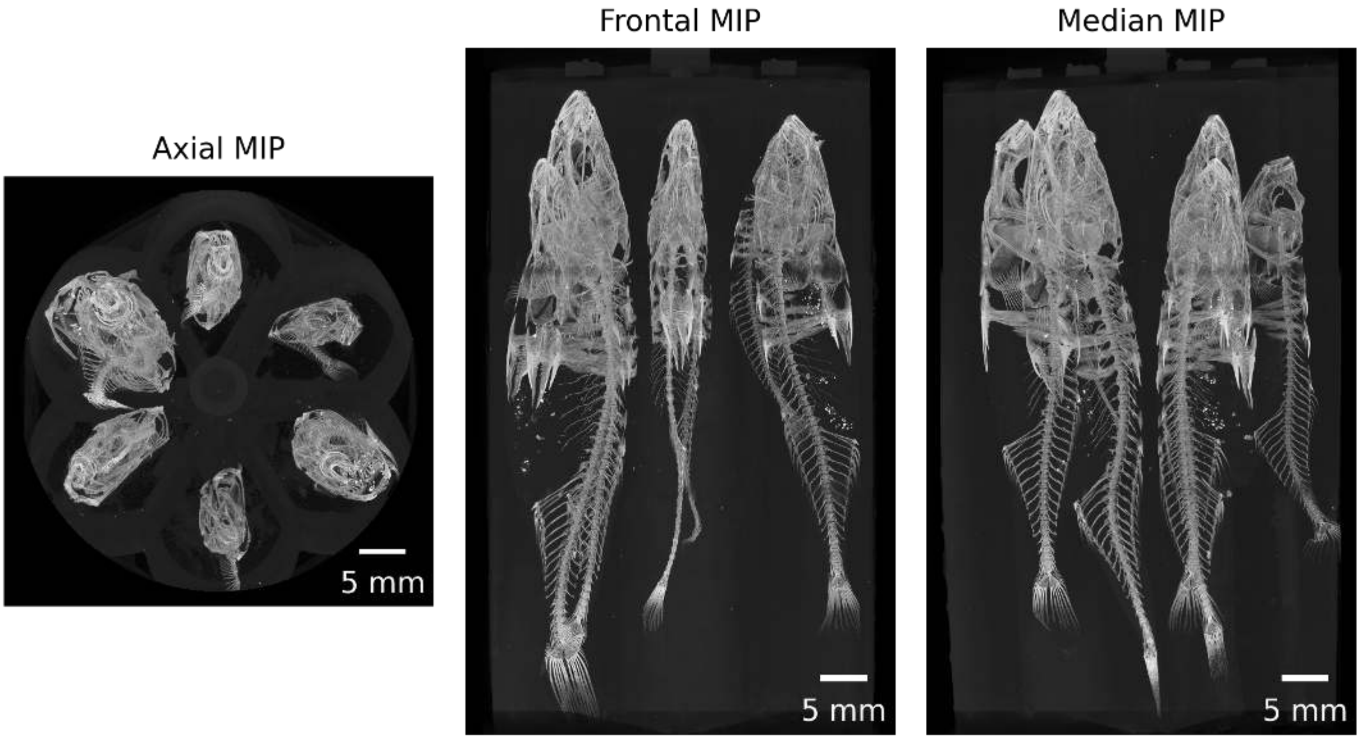

Axial MIP
Frontal MIP
Median MIP
5 mm
5 mm
5 mm

Figure 1: Maximum intensity projections of one acquired dataset along all three cardinal axes.

Separation notebook

The separation notebook processes all the performed scans to extract the single fish out from each scan, where 6 fish have been scanned. As in the preview notebook, we efficiently load all the PNGs from disk with `dask` [6]. Based on the previously extracted MIP images and a simple labeling of the image (`skimage.measure.label`), we extract both the labels in the custom-made sample holder and the positions of single fish in the scan (`skimage.measure.regionprops`) (see Figure 2).

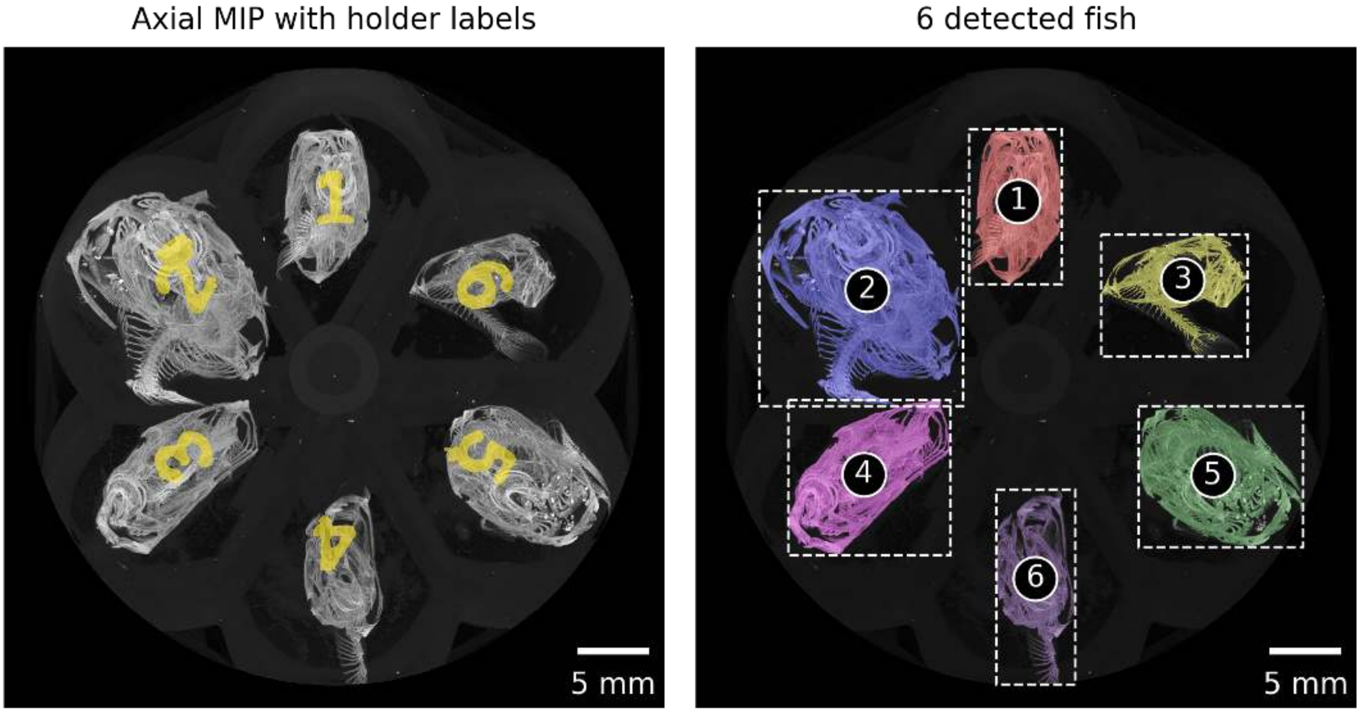
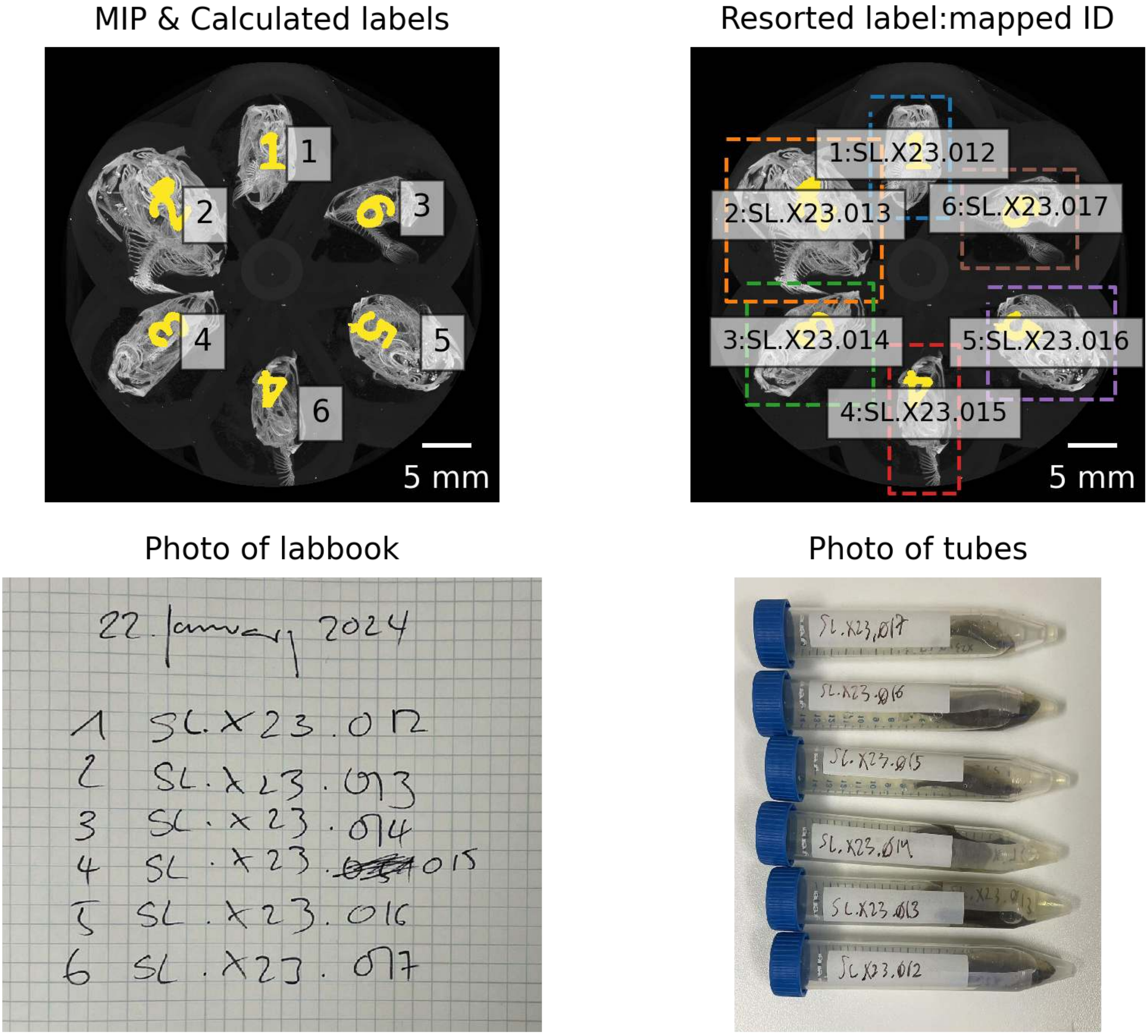

Axial MIP with holder labels
6 detected fish
1
2
3
4
5
6
5 mm
5 mm

Figure 2: Automatically detected regions based on maximum intensity projection along the rotation axis of the tomographic scan. The regions are numbered consecutively from the top left to the bottom right. These numbers are mapped to the correct fish ID in the next step.

Based on a simple mapping of the detected region to the ID numbers of the scanned fish, we labeled the resulting images and presented these images together with photos of the lab book and sample tubes for double-checking (see Figure 3).


MIP & Calculated labels
1
2
3
4
5
6
5 mm
Resorted label:mapped ID
1:SL.X23.012
2:SL.X23.013
6:SL.X23.017
3:SL.X23.014
5:SL.X23.016
4:SL.X23.015
5 mm
Photo of labbook
Photo of tubes
SL.X23.017
SL.X23.016
SL.X23.015
SL.X23.014
SL.X23.013
SL.X23.012

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1 SL.X23.012
2 SL.X23.013
3 SL.X23.014
4 SL.X23.015
5 SL.X23.016
6 SL.X23.017

Figure 3: Mapping lab book notes, photos and detected regions to fish ID.

The `skimage.measure.regionprops` function we used for labeling returns not only the positions of the detected fish, but also the extent of the bounding box of the region of the fish shown in the original image. We extracted each region of each fish separately out of the large reconstructions (with a configurable border buffer, see Figure 4) and wrote these extracted regions to disk in discrete folders for efficient further analysis. In a first step, we wrote the regions of the single fish to disk in `zarr` [7] format, which is a preferred format to store n-dimensional arrays on disk. In addition to this, we also wrote a log file for each extracted region, containing all relevant information to redo the cropping step completely manually (an example of such a log file is shown as part of the processing repository).

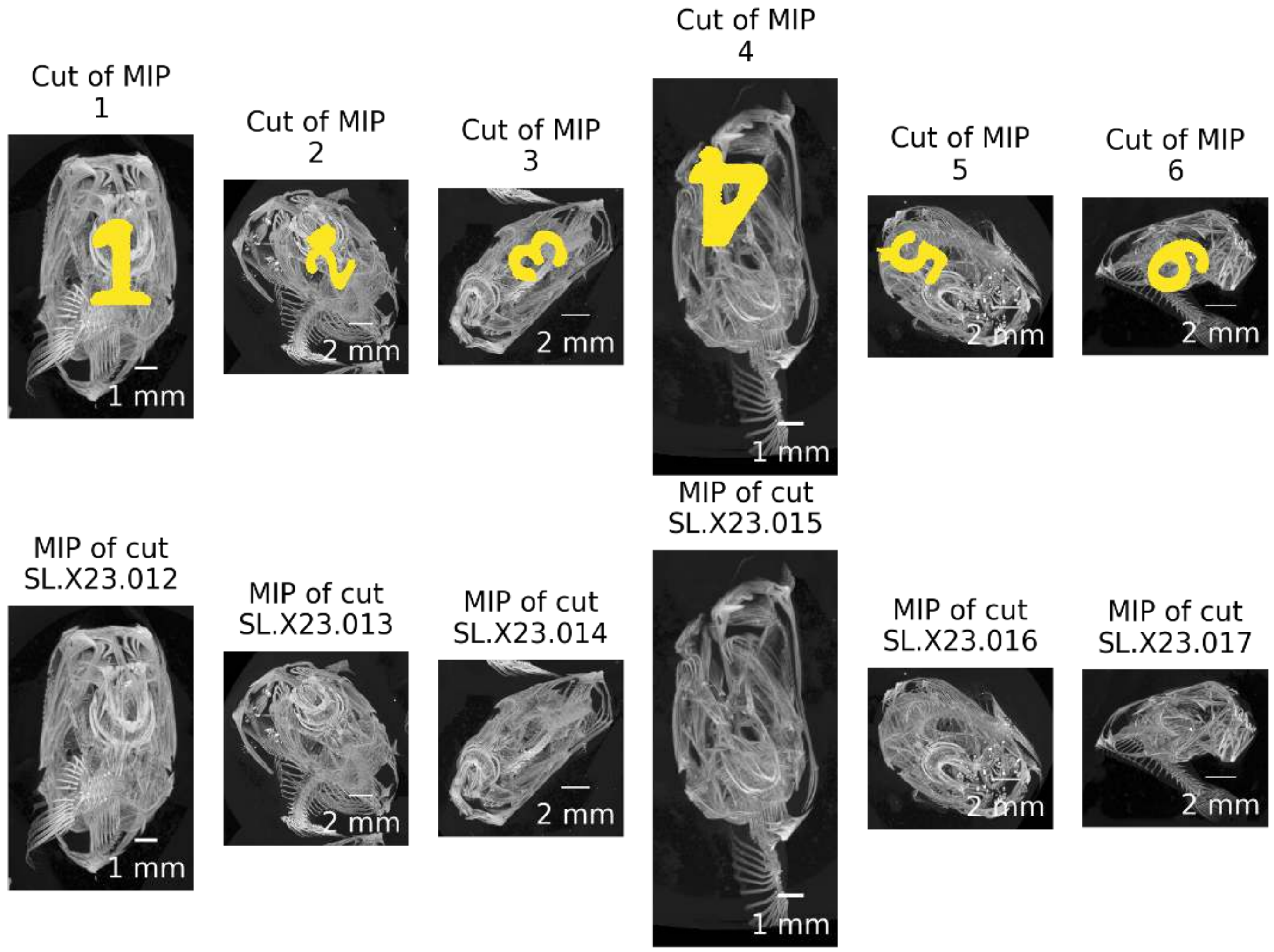
Cut of MIP 1
Cut of MIP 2
Cut of MIP 3
Cut of MIP 4
Cut of MIP 5
Cut of MIP 6
1
2
3
4
5
6
1 mm
2 mm
2 mm
1 mm
2 mm
2 mm
MIP of cut SL.X23.012
MIP of cut SL.X23.013
MIP of cut SL.X23.014
MIP of cut SL.X23.015
MIP of cut SL.X23.016
MIP of cut SL.X23.017
1 mm
2 mm
2 mm
1 mm
2 mm
2 mm

Figure 4: Double-checking crop extent and fish ID.

Saving out the regions as `zarr` files made it possible to efficiently work with the image data of each extracted fish and to convert that data to any desired format for further analysis. For this further analysis, we wrote out stacks of PNG images and additionally, as `nrrd` files for each fish region as a simple crop out of the original dataset and as binarized regions, which are segmented into bone and background based on a simple multi-level Otsu thresholding method [8].

Extraction of features of interest

- Biomedisa [9]

Results

microCT data

- We actually scanned N=216 fishes (unique Fish IDs)
- The total scanning duration was 18 days, 12 hours and 6 minutes
- In total, we acquired 158444 projections
- These were reconstructed into a total of 177749 reconstructions Which is about 4040 files per scan (N=44)

Analysis

- 3D shape variation on internal pharyngobranchial bone. Only possible to get this information in 3D.

Discussion

- Both *repeatable* and *reproducible* research
- Automated cropping out of single fish from combined scan very efficiently uses machine time. Several fish can be scanned together, splitting is performed after the fact, reproducibly and without manual input.
- Combination of methods cuts down on time **a lot**.
- Biomedisa makes more “extraction” possible. Other biological questions can be answered, too.

Conclusion

Author Contributions

Contributor Roles Taxonomy (CRediT), as defined in [10]:

- Conceptualization: Ben Sulser, Ruslan Hlushchuk
- Data curation: David Haberthür, Ben Sulser
- Formal analysis: David Haberthür, Ben Sulser
- Funding acquisition: Ben Sulser, Catherine L. Peichel, Ruslan Hlushchuk
- Investigation: David Haberthür, Ben Sulser
- Methodology: David Haberthür, Ben Sulser, Ruslan Hlushchuk
- Project administration: David Haberthür, Ben Sulser, Catherine L. Peichel, Ruslan Hlushchuk
- Resources: Ben Sulser, Ruslan Hlushchuk
- Software: David Haberthür, Ben Sulser
- Supervision: Ben Sulser, Catherine L. Peichel, Ruslan Hlushchuk
- Validation: David Haberthür, Ben Sulser
- Visualization: David Haberthür, Ben Sulser
- Writing – original draft: David Haberthür, Ben Sulser
- Writing – review & editing: David Haberthür, Ben Sulser, Catherine L. Peichel, Ruslan Hlushchuk

Competing Interest

Author	Competing Interests	Last Reviewed
David Haberthür	None	2026-01-14
Ben Sulser	Nothing to Declare	
Sheila Christen		
Catherine L. Peichel	none	2026-01-19
Ruslan Hlushchuk	None	2026-01-19

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Supplementary Materials

Parameters of tomographic scans of all the fishes

The CSV file ScanningDetails.csv gives a tabular overview of all the (relevant) parameters of all the scans we performed. This file was generated with the data processing notebook and collates the relevant data read from *all* the log files of *all* the scans we performed. A copy of each log file containing *all* scanning parameters is available in a folder in the data processing repository.

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